

Hardware & Networking Basics

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Labs

- Students to setup their virtual lab using Virtual Box.
- They will download the required virtual machines.
- They will setup the virtual network as per the instructions.
- Students will download various machines and access them

Operating Systems



Operating Systems – Windows 11

- Graphical User Interface (GUI) with Start Menu, taskbar, windows, and icons
- Supports a very large range of software applications (Office, browsers, design tools, games, etc.)
- Used widely in homes, offices, schools, and enterprises
- Supports networking, security, updates, and user permissions
- Centered Start Menu
- Requires newer hardware (TPM 2.0, Secure Boot)
- Long-term future



Operating Systems – Mac OS

- This is Apple's operating system used only on MacBooks.
- Key features:
 - ✓ Very smooth, stable, and secure
 - ✓ Excellent for creative work (video editing, music production, graphics)
 - ✓ Works seamlessly with iPhone & iPad
 - ✓ Strong security and privacy
 - ✓ UNIX-based (very powerful backend)



Operating Systems – Android OS

- This is Google's mobile operating system used in:
 - Smartphones
 - Tablets
 - Smart TVs
 - Smart Watches
 - Android Auto (cars)
- Key features:
 - Touch-based interface
 - Highly customizable
 - Massive app support via Google Play Store
 - Open-source (Linux-based)
 - Used by companies like Samsung, Xiaomi, OnePlus, etc.



Golden Images

- A Golden Image is not an operating system itself.
- **It is a perfect, pre-configured master copy of an OS used in organizations.**
- It contains:
 - Operating System (Windows, Linux, etc.)
 - Company-approved software
 - Security updates/patches
 - Drivers
 - Group policies and settings
- It is used for:
 - Quickly deploying multiple identical systems.
 - To ensure standardization and security
 - To save time in large-scale installations (labs, offices, data centers)



Comparison Table

Feature	Windows 10/11	MacOS	Android OS	Golden Image
Type	Desktop OS	Desktop OS	Mobile OS	Master system copy
Developed by	Microsoft	Apple	Google	IT Admin / Org
Devices	PC, Laptop	Mac computers	Phones, Tablets	Used for bulk installation
Customization	Medium–High	Low–Medium	Very High	Fixed configuration
Main Use	Work, gaming, office	Creative, development	Mobile apps	Enterprise deployment

Linux Fundamentals



Basics of UNIX

- Originally came from UNICS (Uniplexed Information and Computing System).
- UNIX is an operating system which was first developed in the 1970s, and has been under constant development ever since.
- It is a stable, multi-user, multi-tasking system for servers, desktops and laptops.
- UNIX systems also have a graphical user interface (GUI) similar to Microsoft Windows which provides an easy-to-use environment.
- There are many different versions of UNIX, although they share common similarities.



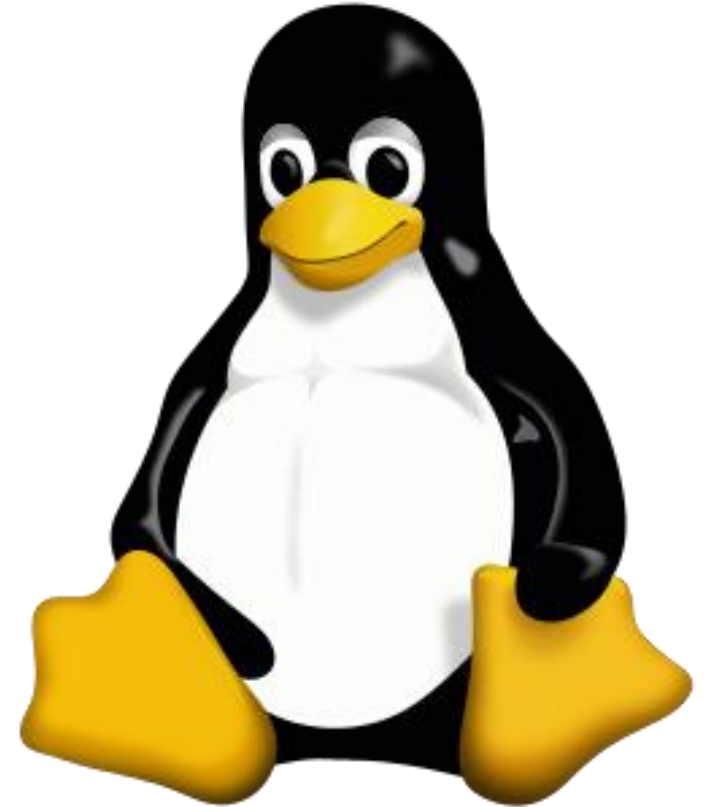
The UNIX Distributions / Distros

- ✓ **AIX** by IBM
- ✓ **BSD**/OS (BSDi) by Wind River
- ✓ **HP-UX** by Hewlett-Packard Company
- ✓ **OpenBSD** by OpenBSD Group
- ✓ **Solaris** by Sun Microsystems (Oracle)
- ✓ **IRIX** by Silicon Graphics, Inc.
- ✓ **Tru64 UNIX**
- ✓ **FreeBSD**
- ✓ **NetBSD**
- ✓ **Mac OS**



Linux

- Linux is a kernel, not a complete operating system.
- The kernel is the core part of an operating system.
- It manages:
 - CPU
 - Memory (RAM)
 - Devices (keyboard, disk, etc.)
 - Processes
- Linux by itself cannot work for normal users without other software.



Why to choose LINUX ?

- Easy to install applications
- Secure
- Easy to change options → Graphical Machines (GUI)
- Community
- Free
- Portable
- Open Source
- Multi-User Hierarchical File System (FHS)



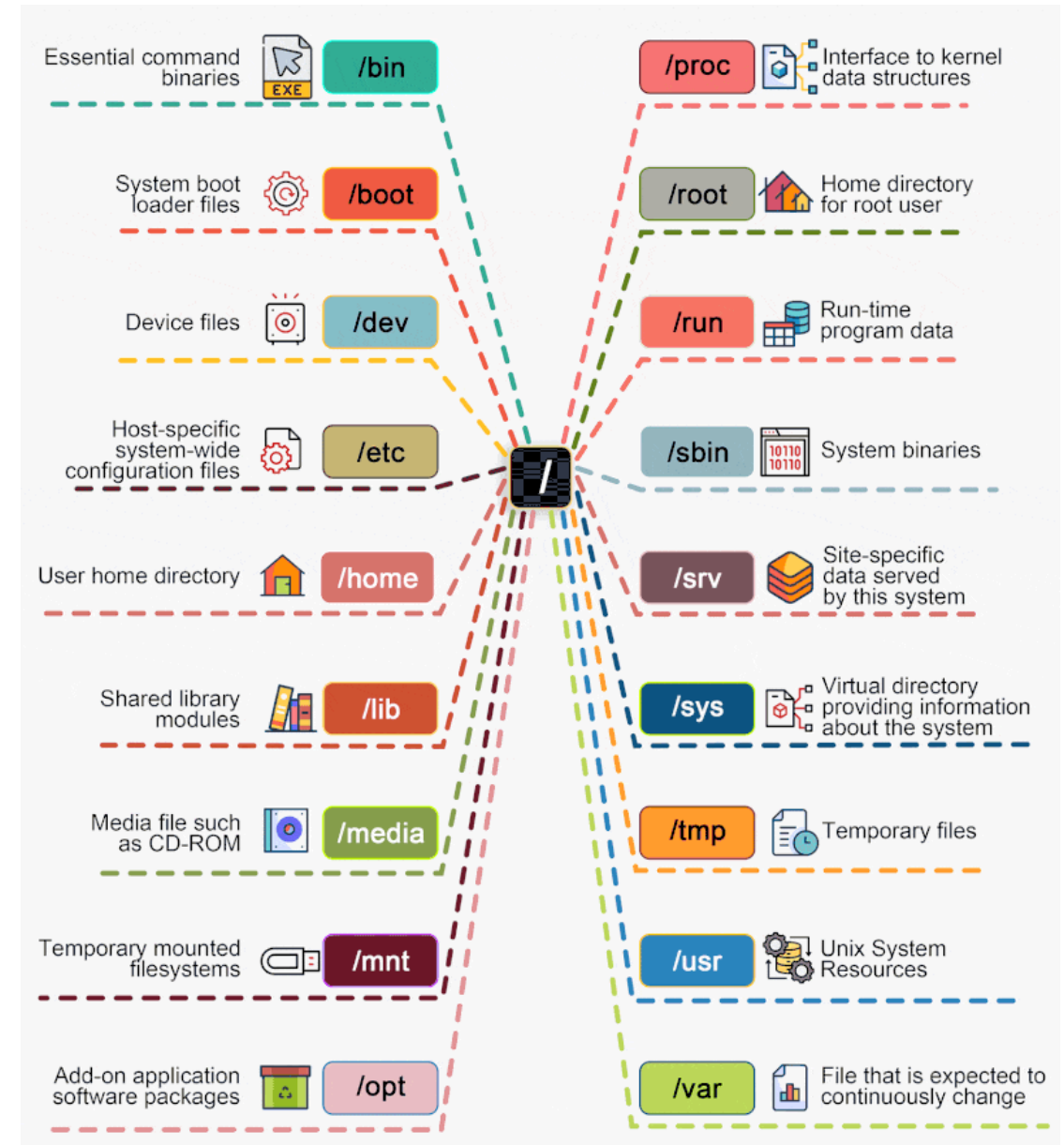
Linux Distributions

- Ubuntu
- Linux Mint
- Debian
- Fedora
- OpenSUSE / SUSE Linux Enterprise
- Mandriva
- Puppy Linux
- Slackware Linux
- Parrot linux
- Kali linux
- Back Track
- Arch Linux
- Red Hat Enterprise Linux / CentOS



Linux File System

- The Linux File System is a tree-like (hierarchical) structure that starts from a single root directory (/).
- Linux doesn't use drive letters like C:, D.
- Types of Linux File Systems:
 - EXT4 → Most common
 - XFS → Enterprise servers
 - BTRFS → Advanced features
 - SWAP → Virtual memory



Kernel

It is responsible for:

- **Process Management**

- Creates, schedules, and terminates processes
- Manages CPU time

- **Memory Management**

- Allocates and deallocates RAM
- Handles virtual memory & paging

- **Device Management**

- Communicates with hardware via drivers
- Controls keyboard, disk, USB, network card, etc.

- **File System Management**

- Handles reading/writing data to storage
- Manages file permissions and access

- **Security & Access Control**

- Enforces user permissions
- Protects system memory from misuse

- **Networking**

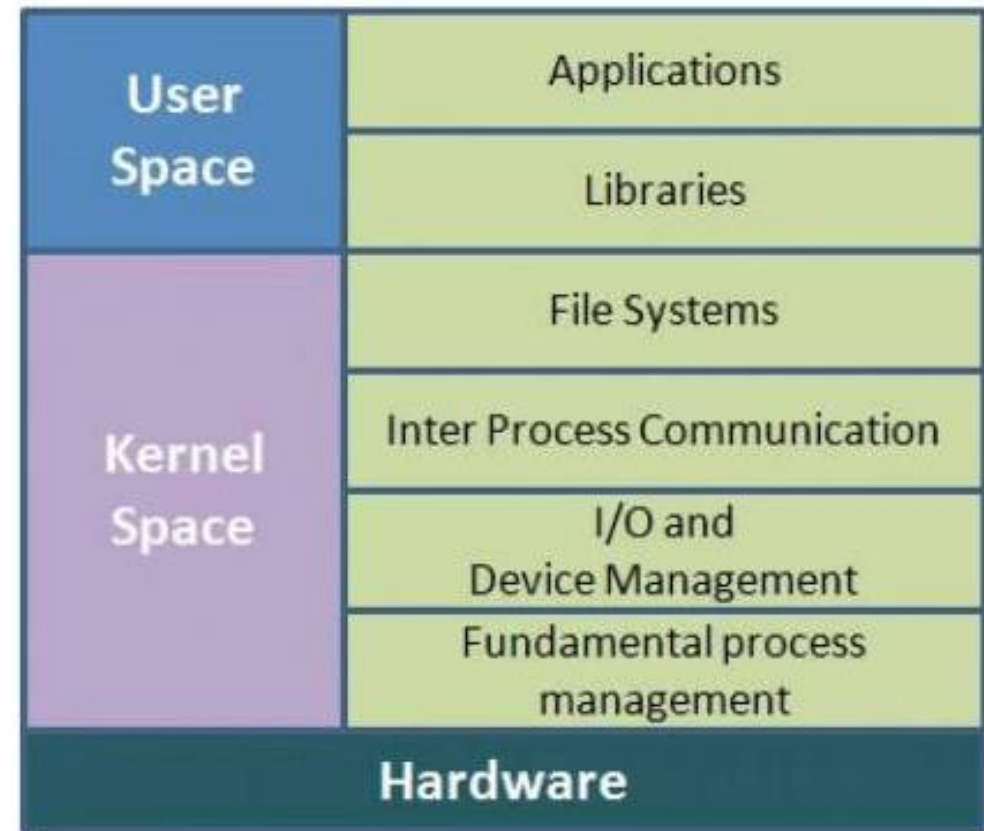
- Handles data transmission over network

Types of Kernels

Type	Description	Example
Monolithic kernel	All services run in kernel space	Linux
Microkernel	Minimal function in kernel	Minix
Hybrid kernel	Mix of both	Windows, macOS

Monolithic Kernel

- A Monolithic Kernel is a kernel design where all core operating system services run in kernel space as a single large program.
- That includes:
 - Process management
 - Memory management
 - File system
 - Device drivers
 - Network stack
 - System calls
- Everything runs with full system privileges.



Monolithic Kernel

- **Advantages**

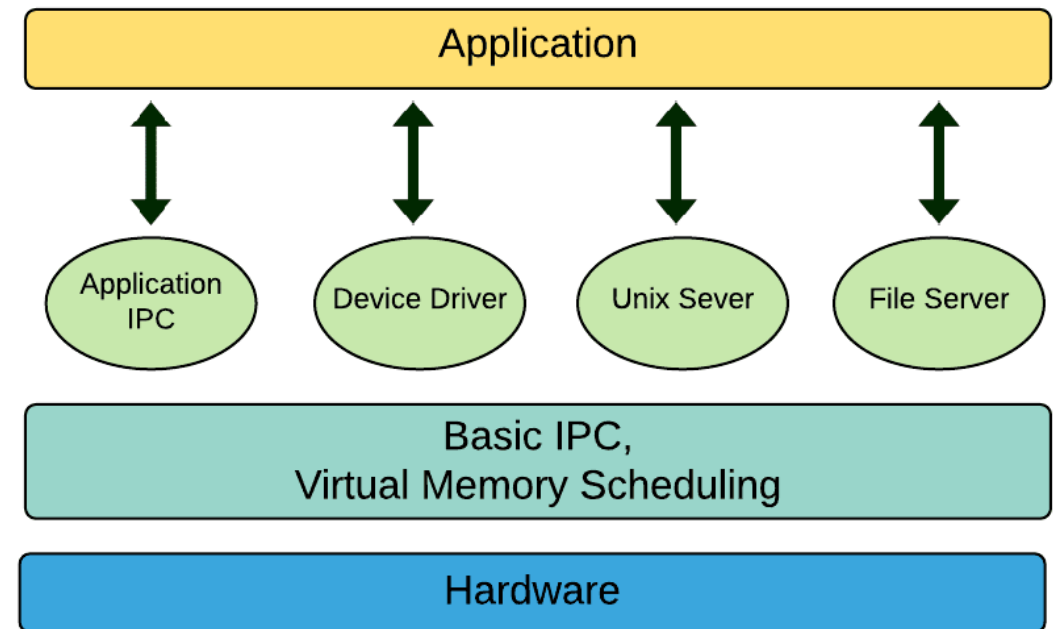
- Very fast
- Less communication overhead
- High performance
- Mature & well-optimized (Linux)

- **Disadvantages**

- Large and complex
- A bug in any part can crash the entire system
- Harder to maintain

Microkernel Kernel

- A Microkernel is designed to keep the kernel as small as possible, and runs most services in user space instead of kernel space.
- Only essential tasks run in kernel mode:
 - Inter-process communication (IPC)
 - Basic scheduling
 - Basic memory management
- Other services (drivers, file system, network) run in user mode.



Microkernel Kernel

- **Advantages**

- More stable (crash in one service won't crash whole system)
- More secure
- Easy to maintain and modify
- Better modularity

- **Disadvantages**

- Slower than monolithic (communication overhead)
- More complex message passing
- Performance cost

Daemons in Linux

- Daemons are background service processes in Linux that run continuously without direct user interaction.
- They usually start at system boot and keep running to provide various services.
- Key Characteristics of Daemons
 - Run in the background
 - Usually start at boot time
 - Do not need a user interface
 - Always listening or waiting for an event
 - Typically end with the letter “d”



Daemons Examples:

- **sshd** – Secure Shell (SSH)
- **httpd** – Web server (Apache)
- **crond** – Scheduled job manager
- **systemd** – Main system manager
- **mysqld** – MySQL database
- **cupsd** – Printer service
- Modern system command to check using systemctl

```
# systemctl list-units --type=service
```






That's all Folks!